

PROFILE

Robotics and Control Engineer (PhD, TU Delft) with expertise in dynamics, motion planning, fault-tolerant control, and ROS2-based system integration. Strong mechanical engineering foundation (kinematics, dynamics, mechanisms, design, CAD/CAE) and hands-on deployment of autonomous platforms (ASVs, UGVs). Solid background in dynamical systems, control, estimation, and optimization, with experience translating advanced research into validated engineering solutions in safety-critical environments. Currently working on real-world deployment of maritime autonomy systems in an industrial R&D setting.

CORE SKILLS & COMPETENCIES

- **Robotics software & system integration:** ROS 2 (simulation to deployment), sensor/actuator integration, system validation and testing, model-based systems engineering principles
- **Motion planning & control:** Real-time NMPC, state estimation, trajectory optimization, fault diagnosis, fault-tolerant and safety-critical autonomous systems
- **Programming:** C++ (real-time, mission-level control), Python, Linux, MATLAB/Simulink
- **Engineering tools:** Git; CAD (SolidWorks, Inventor, Creo Parametric); FEA/CAE (ANSA, Altair MotionView)
- **Languages:** Greek (native), English (fluent), French (A1)

EXPERIENCE

R&D Engineer (Autonomous Systems)

Apr 2026 – Present

DeepSea, Greece

- Developing autonomy and decision-support algorithms for maritime applications, focusing on motion planning, control, and system integration.
- Collaborating with cross-functional teams and international partners to integrate autonomy modules into real-world vessel operations.

Independent Engineering Projects (listed below)

Jun 2025 - Apr 2026

Personal Projects

- Ongoing development of robotics and control software with focus on system integration, testing, and operational readiness.
- Hands-on work on autonomous platforms, sensor integration, real-time control, and validation workflows.

Research & Development (PhD)

Dec 2020 - Jun 2025

TU Delft, The Netherlands

- Designed and validated real-time motion planning and control algorithms enabling safe autonomous navigation under faults, uncertainty, and operational constraints.
- Implemented and integrated ROS-based planning and control pipelines, extensively tested in simulation and prepared for real-world deployment on autonomous platforms.
- Developed automated testing, logging, and performance evaluation workflows for systematic validation and reproducibility.
- Provided technical supervision to MSc students on control design, software architecture, experiments, and validation.

Teaching Assistant

Nov 2019 - Jan 2020

TU Delft, The Netherlands

- Supported MSc-level course on nonlinear systems theory, guiding and evaluating students in assignments.

Mechanical Engineering Intern

Jul 2016 - Oct 2016

Maniatopoulos S.A. IDEAL BIKES, Patras, Greece

- Designed and prototyped components and tools for a cargo e-bike.
- Performed statistical quality analysis to reduce frame defects and improve manufacturing robustness.

Formula Student Team Member

Jul 2013 - Oct 2014

Aristotle Racing Team, Thessaloniki, Greece

- Led design, structural analysis (CAD/CAE), and manufacturing of the vehicle's steel tubular frame.
- Coordinated subsystem integration, subsystem interfaces, assembly sequencing, rule-compliance, and validation.
- Ensured competition readiness; achieved top-10 placements internationally.

EDUCATION

PhD in Cognitive Robotics

Dec 2020 – Jun 2025

TU Delft, The Netherlands

- Thesis: *"Rule-compliant and Fault-tolerant Motion Planning: With Application to Autonomous Surface Vehicles"*.
- Developed real-time algorithms for collision avoidance with rule compliance, fault diagnosis, fault reconfiguration, and robustness to uncertainties, resulting in safer and more reliable navigation compared to existing solutions.
- Collaborated within an international, interdisciplinary research team and co-supervised MSc students on projects related to autonomous navigation.

MSc in Systems and Control

Sep 2018 – Dec 2020

TU Delft, The Netherlands

GPA: 8.56/10 **Cum Laude**

- Thesis: *"Distributed IDA-PBC for Nonholonomic Mechanical Systems"*. Designed a distributed passivity-based control algorithm for a fleet of heterogeneous, underactuated, and nonholonomic robots.
- Specialized in control theory, dynamical systems, optimization, and adaptive/predictive/robust control methods.
- Collaborated in small international teams to deliver high-quality results in demanding course projects.

M.Eng. in Mechanical Engineering (Integrated BSc & MSc equivalent)

Sep 2011 – Jul 2018

Aristotle University of Thessaloniki, Greece

GPA: 8.08/10

- Thesis: *"The Effect of Power Distribution Architectures and Torque Vectoring in Vehicle Dynamics"*. Investigated vehicle dynamics under different power distribution architectures and developed a torque-vectoring controller.
- Specialized in mechanical design, analysis, synthesis, and control of mechanical systems.
- Member of Formula Student team; collaborated to deliver a competitive racing vehicle for international competitions.

AWARDS & DISTINCTIONS

- MSc Systems & Control *Cum Laude*, TU Delft.
- Young Author Award nominations (IFAC-MICNON 2021, IFAC-CAMS 2022).
- Formula Student: 4th overall (FS Italy); 12th overall (FS Germany); top-10 in Engineering Design (FS Hungary).

PERSONAL PROJECTS

- **nav-mpc — Real-time Nonlinear MPC Framework for Autonomous Navigation** Nov 2025 – Present
Nonlinear MPC · OSQP · Embedded Control · Python · ROS2
 - Built a real-time NMPC framework with automatic reformulation from symbolic NMPC to LTV-QP reformulation with fast Cython-accelerated evaluation.
 - Real-time QP solution using OSQP, achieving sub-millisecond solution times.
 - Integrated simulation, logging, and benchmarking tools to support systematic validation and reproducible testing.
 - Demonstrated on dynamic systems and mobile robots; designed for embedded use on UGV/UAV/USV platforms.
- **Autonomous Rover — ROS2-Based Mobile Robot with Full Onboard Autonomy** Sep 2025 – Present
Rover · Raspberry Pi 5 · LiDAR · IMU · Odometry · ROS2
 - Designed and assembled an autonomous rover platform; performed hardware/software bring-up, 3D-printed parts, custom mounting, cable management, logging, and field-run analysis.
 - Implemented EKF-based state estimation from wheel odometry, IMU, and 2D LiDAR; prepared for closed-loop autonomy tests.
 - Modular onboard architecture separating perception/SLAM and planning/control for reliable real-time operation.
- **FLARE — Solar-Powered UAV for Persistent Wildfire Surveillance** Sep 2025 – Present
UAVs · Solar Power · Environmental Monitoring
 - Developed a wildfire spread simulator using cellular automata and GeoTIFF-based vegetation maps.
 - Conducted payload studies and preliminary airframe sizing and optimization for > 24 h endurance.
 - Conceptual design of autonomy stack for a solar-powered autonomous glider with thermal and RGB payloads.
 - Following a model-based systems engineering (MBSE) approach using the V-model, stakeholder analysis, operational scenarios, functional analysis, and requirements definition.
 - Submitted as an MSCA Postdoctoral Fellowship (MSCA-PF) research proposal (scored 82%).

PUBLICATIONS (SELECTED)

- **Tsolakis, A.**, Negenborn, R., Reppa, V., Ferranti, L. *Model Predictive Trajectory Optimization and Control for Autonomous Surface Vessels Considering Traffic Rules*. IEEE Trans. Intell. Transp. Syst., 25(8):9895–9908.
- **Tsolakis, A.**, Ferranti, L., Reppa, V. *Set-Membership Estimation for Fault Diagnosis of Nonlinear Systems*. In *Proc. European Control Conference (ECC)*, Thessaloniki, Greece, 24–27 Jun. 2025.
- **Tsolakis, A.**, Benders, D., De Groot, O., Negenborn, R., Reppa, V., Ferranti, L. *COLREGs-aware Trajectory Optimization for Autonomous Surface Vessels*. IFAC-PapersOnLine 55(31):269–274.
- **Tsolakis, A.**, Keviczky, T. *Distributed IDA-PBC for a Class of Nonholonomic Mechanical Systems*. IFAC-PapersOnLine 54(14):275–280.